Lab 2 – FreeRTOS "Hello World!"

Requirements

1. Refactor your solution to Lab 1 where:
    a. Your name is displayed once on the UART port each time that "User B1" button is pressed (it is not displayed at power-up and it is not displayed continuously while the button is held pressed)
    b. The GREEN LED flashes at precisely 1 Hz
2. Your solution must be composed of two independent FreeRTOS tasks
3. You must demonstrate proper operation of your board during the lab time
4. You must submit your source code (only the source code, and only the files that you modified or created).

Hints

- You may want to enable **`vTaskDelayUntil()`** in FreeRTOS
- Do not try to copy your previous project, create a new one within CubeMX
- Check the "Enable multi-threaded support" checkbox within CubeMX
- Select CMSIS_V2 for the interface
- The GPIO pin for the push button is defined in the CubeMX project (not in your book)

Assessment Criteria

- Name displayed when "User B1" is pressed
    - 2pts
- GREEN LED flashes at precisely 1 Hz
    - 1pt – LED Flashes
    - 2pts – Precisely at 1 Hz
- Two independent FreeRTOS tasks, one for each required function
    - 1pts – FreeRTOS initialization / task creation
    - 2pts – Tasks
- 2pts – Code quality (readability, comments, structure, etc.)
- 3pts deducted for each day that the demo is late

Objective

The purpose of this lab is to get you familiar with creating a FreeRTOS embedded application composed of multiple tasks that do not interact with each other in any way.